Retraining Matters More Than the Model: Day-Ahead Price Forecasting for Germany with Grid-Operator Forecasts

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Abstract

German day-ahead electricity prices are increasingly driven by wind and solar generation, which makes them a natural test case for comparing time-series models with supervised models that use forecasts of the physical drivers. Such comparisons are often made with observed instead of forecast weather, or with a model trained once and evaluated over a period in which the market changes. We compare naïve baselines, Prophet, Ridge regression and gradient-boosted trees on the year from October 2025 to October 2026 under a strict day-ahead information set, using the grid operators' day-ahead forecasts of load, wind and solar generation and treating observed weather only as an upper bound. Gradient boosting with lagged prices and these forecasts, expressed as shares of the forecast load and retrained every two weeks on a rolling 180-day window, reaches a mean absolute error of 15.98 €/MWh, 48.6 % lower than repeating yesterday's price ($p < 0.001$) and lower in every month. Any window between 90 and 270 days gives the same result within the uncertainty of a paired bootstrap (maximum paired difference to the 180-day window of 0.24 €/MWh). Trained once, the same model is only 18 % better and falls behind the naïve forecast from July 2026; a drift analysis shows that the price response to residual load shifted while the inputs did not. Retraining also reverses the ranking of inputs and algorithms: observed weather, absolute forecasts and Ridge regression win under a single fit and lose with retraining, after which perfect weather knowledge adds at most 0.3 €/MWh. TreeSHAP explanations recover a seasonal merit-order curve. Which model ranks highest depends on the training scheme; the training scheme itself is the first-order decision. A supervised model needs time-series memory in its inputs and time-series practice in its training.

1 Introduction

German day-ahead electricity prices are increasingly set by the weather. Wind and solar generation have near-zero marginal cost, so each additional megawatt-hour pushes more expensive thermal plants out of the merit order and lowers the clearing price [1, 2]. As their capacity has grown, this effect has reshaped the daily price curve. Middays with strong solar generation are often cheap or negative, the share of negative-price hours rose from 3.4 % in 2023 to 6.5 % in 2025, and prices climb steeply in the evening when solar output fades and flexible thermal plants are needed. On top of this, the price level is still shaped by fuel and carbon prices, which moved sharply during and after the 2021–2022 gas crisis. Accurate day-ahead forecasts matter for anyone who has to commit before the auction: storage operators, flexible consumers, and traders.

Two families of models compete for this task. Time-series models, from the naïve "same hour yesterday" forecast to seasonal models such as Prophet, use the persistence and the daily, weekly and yearly cycles of the price itself, but they ignore its main physical driver. Supervised models can be given forecasts of load, wind and solar generation and learn how the price responds to them, but on their own these inputs do not tell the model where the current price level is. Which

family is better suited to a market like Germany's is not obvious, and the answer depends on how the comparison is made. Published comparisons often give the models information that a forecaster would not have, such as observed instead of forecast weather, or train a model once and evaluate it over a long period in which the market changes.

This study asks the question under conditions that a forecaster would actually face. Under a strict day-ahead information set and an evaluation design without look-ahead, can a supervised model beat strong naïve baselines on German day-ahead prices, and what combination of lagged prices, realistic grid-operator forecasts and retraining is required?

To answer it, we compare naïve baselines, Prophet, Ridge regression and gradient-boosted trees on one fixed test year, 29 October 2025 to 28 October 2026, with the information available at the day-ahead auction. The inputs are the day-ahead forecasts of load, wind and solar generation published by the German transmission system operators (TSOs). Observed weather is used only as an upper bound. The experiments proceed in stages, from single fits to walk-forward retraining with different training windows, and every comparison is tested with the Diebold–Mariano test.

The study makes four contributions. First, it defines a reproducible evaluation protocol for day-ahead price forecasting that removes look-ahead from inputs, training and preprocessing, and that separates realistic inputs from oracle inputs. Second, it shows that gradient boosting with lagged prices, TSO forecasts expressed as shares of the forecast load, and walk-forward retraining on a rolling 180-day window reaches an MAE of 15.98 €/MWh, 48.6 % better than the 24 h naïve forecast, significantly better and better in every month of the test year. Any window between 90 and 270 days gives practically the same result, so the headline does not depend on the window that was chosen. Third, it shows that the training scheme decides the ranking of models and inputs: observed weather, absolute instead of relative TSO forecasts, and Ridge regression all look better than the alternatives under a single fit, and all lose once the models are retrained. With retraining, perfect knowledge of the weather is worth at most about 0.3 €/MWh on top of the TSO forecasts. Fourth, it diagnoses why a single fit fails. The degradation is explained by a shift in how the price responds to the residual load, an upward move of the learned merit order, and not by a shift in the inputs. Exact TreeSHAP explanations of the retrained models show the merit-order curve and how its shape and the importance of each input change with the season.

Section 2 reviews related work. Section 3 describes the data and the experimental design, section 4 presents the results, section 5 discusses them together with the drift diagnosis and the limitations, and section 6 concludes.

2 Related work

Electricity price forecasting. Day-ahead electricity price forecasting has a large literature, reviewed by Weron [3] for point forecasts and by Nowotarski and Weron [4] for probabilistic forecasts. Lago et al. [5] reviewed recent algorithms, proposed best practices and published an open benchmark. Their recommendations include test periods of at least one year, comparison against simple and well-established benchmarks, statistical tests such as Diebold–Mariano, and models that are recalibrated as new data arrive; their benchmark datasets use day-ahead forecasts of load and renewable generation as exogenous inputs. Ziel and Weron [6] studied high-dimensional linear models with many lagged prices, estimated with the lasso, which remain strong benchmarks. The present study follows these practices and adds two elements they leave open: an explicit oracle upper bound for weather information, and a diagnosis of why models degrade without recalibration.

Renewables and the merit order. The price-lowering effect of wind and solar generation on the German spot market is well documented. Sensfuß et al. [1] quantified this merit-order

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effect for Germany, and Hirth [2] showed how the value of wind and solar falls as their share rises. Both explain why residual load, the load not covered by wind and solar, is a natural input for price models, and why the relationship between residual load and price is curved rather than linear.

Weather inputs and evaluation practice. Several recent studies predict prices from weather and renewable-generation variables with supervised learning. Debroy et al. [7] compare linear regression, random forests and XGBoost on German prices from 2016 to 2024 without time-series baselines, and Zhao et al. [8] compare eight algorithms on the Spanish market from 2015 to 2018 with a random 80/20 split, which mixes past and future observations in training and test data. The closest recent work is Hertel et al. [9], who forecast German day-ahead prices with a Transformer designed for fast SHAP explanations. Their study and ours share the market, the period and the use of SHAP, but differ in the evaluation. They forecast 168 hours ahead with ERA5 reanalysis weather, which is observed rather than forecast weather, train once on data up to 2023 and test on 2025, and compare against a weekly persistence forecast and a default XGBoost model with lag-168 prices only. We use the TSO day-ahead forecasts as realistic inputs, measure what observed weather adds as an upper bound, retrain the models through the test year, and compare against the strongest naïve forecast we found.

Calibration windows and concept drift. How much history to train on is a known question in price forecasting. Marcjasz et al. [10] showed that the length of the calibration window affects accuracy and that combining forecasts from short and long windows can beat any single window, and Hubicka et al. [11] found similar gains from averaging across windows. More generally, models deployed on data whose distribution changes over time face concept drift, for which Gama et al. [12] give an overview of detection and adaptation methods, including retraining on sliding windows. We add a measured window sensitivity for the post-crisis German market, with a plateau and a step confirmed by paired bootstrap intervals, and a diagnosis that separates shifts in the inputs from shifts in the price response.

Positioning. The present study combines three elements that are rarely examined together: realistic day-ahead forecasts instead of observed weather, the price level through lagged prices, and walk-forward evaluation over a year in which the market changed. Its main finding is that the third element changes the conclusions drawn about the first two.

3 Data and experimental design

All series are hourly and are converted to coordinated universal time (UTC) before they are joined. The modelling window starts on 1 January 2023; longer histories are used only for descriptive statistics and to compute lagged features at the start of the window.

3.1 Day-ahead prices

The target is the hourly day-ahead price of the German–Luxembourg bidding zone (DE-LU), as cleared in the European single day-ahead coupling auction operated through EPEX Spot. Prices were retrieved from the public market-data API of aWATTar [13] for 1 October 2016 to 28 October 2026 (88,320 hours, no missing values). Before 1 October 2018 the bidding zone also included Austria (DE-AT-LU); this period lies outside the modelling window. From delivery day 1 October 2025 the auction clears 15-minute products; the hourly series then most likely corresponds to the official 60-minute index, the arithmetic average of the four quarter-hour prices [14] (section 5.5).

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Table 1 summarises the series by calendar year. Three regimes are visible: a low-price period up to 2020 (annual means of 30–45 €/MWh), the gas-price crisis of 2021–2022 (mean 235.4 €/MWhin 2022), and the post-crisis market from 2023 on. The post-crisis years combine a higher price level than before the crisis with a steadily rising share of negative-price hours (from 3.4 % in 2023 to 6.4–6.5 % in 2025–2026) and occasional extreme prices, including a maximum of 2,325.8 €/MWhin 2024.

Table 1: Hourly DE-LU day-ahead prices by calendar year. 2016 starts on 1 October and 2026 ends on 28 October. Values in €/MWh except the share of negative-price hours.

Year	Mean	Std. dev.	Max	Negative hours (%)
2016	37.6	15.6	105.0	2.2
2017	34.2	17.7	163.5	1.7
2018	44.5	17.8	128.3	1.5
2019	37.7	15.5	121.5	2.4
2020	30.5	17.5	200.0	3.4
2021	96.8	73.7	620.0	1.6
2022	235.4	142.8	871.0	0.8
2023	95.2	47.6	524.3	3.4
2024	79.6	64.5	2,325.8	5.2
2025	89.3	52.1	583.4	6.5
2026	112.4	68.2	697.3	6.4

The modelling window starts after the crisis because a price level of several hundred €/MWh teaches a model relationships that no longer hold. In preliminary runs, a Prophet model trained on ten years of data was worse than the same model trained from 2023 (section 4).

3.2 Grid-operator day-ahead forecasts

The exogenous inputs are the day-ahead forecasts that the German transmission system operators (TSOs) publish for the following day: generation from onshore wind, offshore wind and photovoltaics, total grid load, and residual load (grid load minus wind and solar generation). They were downloaded from SMARD, the market-data platform of the Bundesnetzagentur [15], for the DE-LU bidding zone at hourly resolution from 1 October 2018 to 6 November 2026. In the downloaded files, residual load equals grid load minus the three renewable forecasts to rounding precision.

Time zone. SMARD timestamps are local German time, so each year contains one missing hour (spring clock change) and one repeated hour (autumn clock change). The timestamps were localised to the Europe/Berlin time zone, with the repeated hour resolved by order, and converted to UTC. The resulting series has no gaps or duplicates, and the forecast solar generation peaks at 11:00 UTC, as expected for Germany.

Missing values. Gaps of up to three hours are filled by linear interpolation. After that, 126 hourly values are still missing within the modelling window, all of them in 2023; the test year contains none. The gradient-boosting models handle these missing values natively, and the Ridge models impute them with the training-set median (section 3.6).

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Ratios instead of absolute values. Besides the forecasts in gigawatts, four ratios to the forecast grid load L_t are used:

$$s_t^{\mathrm{RL}} = \frac{L_t - W_t^{\mathrm{on}} - W_t^{\mathrm{off}} - S_t}{L_t}, \qquad s_t^{\mathrm{PV}} = \frac{S_t}{L_t}, \tag{1}$$

$$s_t^{\mathrm{W}} = \frac{W_t^{\mathrm{on}} + W_t^{\mathrm{off}}}{L_t}, \qquad s_t^{\mathrm{RE}} = \frac{W_t^{\mathrm{on}} + W_t^{\mathrm{off}} + S_t}{L_t} = 1 - s_t^{\mathrm{RL}}, \tag{2}$$

where W_t^{on}, W_t^{off} and S_t are the forecast onshore wind, offshore wind and solar generation. The ratios describe the position on the merit order more directly than absolute values: 20 GW of wind on a summer Sunday with low demand is a surplus, while the same 20 GW on a winter weekday covers only part of the load. A residual-load share below zero means that the forecast wind and solar generation exceed the forecast load.

Redundant ratios. Since $s_t^{\mathrm{RE}} = 1 - s_t^{\mathrm{RL}}$, the two features carry the same information. Both were kept, as in all experiments. For the predictions this is harmless: tree models simply split on either one, and the L2 penalty of the Ridge model gives a unique solution that divides the weight between them. It matters for the explanations, because TreeSHAP divides the credit between the two features arbitrarily. Their SHAP contributions are therefore summed and reported as a single residual-load effect (section 4.5). As a check, the main model was refitted without s^{RE}: the MAE changed from 15.98 to 15.87 €/MWh, a difference that is not significant (Diebold–Mariano $p = 0.23$).

Publication time. The TSOs are required to report their day-ahead forecasts to the ENTSO-E Transparency Platform by 18:00 on the day before delivery [16], which is after the noon gate closure of the day-ahead auction. SMARD does not expose per-forecast publication timestamps, so we cannot verify the noon cutoff empirically. The forecasts are produced earlier and are the standard exogenous inputs in day-ahead price forecasting studies [5]; we therefore treat them as available before the auction while noting the formal 18:00 reporting deadline as a limitation (section 5).

3.3 Observed weather (upper bound only)

To measure what perfect knowledge of the weather would add, observed hourly weather was retrieved through the Bright Sky API [17], which serves the hourly station observations of the German Weather Service (DWD) [18]. For each of five locations chosen to represent the main German generation and demand regions, Helgoland (North Sea, 54.18 °N, 7.89 °E), Bavaria (48.37 °N, 10.90 °E), central Germany (51.16 °N, 10.45 °E), North Rhine-Westphalia (51.22 °N, 6.77 °E) and Berlin/Brandenburg (52.52 °N, 13.40 °E), Bright Sky returns the observations of the nearest DWD station within 50 km and fills gaps from neighbouring stations. The variables are wind speed at 10 m (mean over the preceding hour), air temperature at 2 m (instantaneous) and solar irradiation (accumulated over the preceding hour, kWh/m^2). They are condensed into national means plus the per-location wind and solar columns. Sunshine duration was excluded because about a quarter of its values are missing before 2023. Wind speed at 10 m is well below the hub height of modern turbines, which is one reason why observed weather is a weaker input than the TSO generation forecasts.

Because these are observations rather than forecasts, models that use them have information no forecaster has at the time of the auction. They are labelled *oracle* throughout and serve only as an upper bound for the value of weather information.

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3.4 Forecast task and information set

The task mirrors the day-ahead auction: at noon on day $D-1$, forecast the 24 hourly prices of day D. All prices of day $D-1$ are known at that time, because they were published on day $D-2$. Price features therefore use only lags of at least 24 hours. Table 2 lists the feature groups.

No look-ahead. Four rules keep future information out of every forecast. (i) Every price feature for an hour of day D uses prices of day $D-1$ or earlier. (ii) Calendar features are known in advance. (iii) The TSO forecasts for day D are issued on day $D-1$; they are forecasts, not measurements, subject to the publication-time caveat in section 3.2. (iv) Every model, and every preprocessing step that is fitted to data (imputation and standardisation in the Ridge pipeline), is fitted only on hours before the block it forecasts. The only exception is the linear interpolation of short gaps in the TSO series, which uses the neighbouring hours; it affects 2023 only, which is training data. The observed weather is the one deliberate exception to these rules and is labelled oracle wherever it is used.

Table 2: Feature groups. All price features use information available by noon on $D-1$.

Group	Count	Features
Calendar	4	hour of day (UTC), day of week, month, German public holiday (0/1)
Price level	7	price 24 h, 48 h and 168 h earlier; mean of the same hour over the previous 7 days; mean, minimum and maximum of the previous day
TSO ratios	4	s^{RL}, s^{PV}, s^{W}, s^{RE}
TSO absolute	6	forecast load, residual load, onshore wind, offshore wind, solar (GW), and s^{RE}
Weather (oracle)	14	national mean wind speed, solar radiation and temperature; Helgoland wind speed; per-location solar radiation and wind speed

3.5 Training and test protocol

The test period is fixed at the last 365 days of the price series, from 29 October 2025 00:00 to 28 October 2026 23:00 UTC (8,760 hours), and is the same for every model. Two training schemes are compared:

Single fit. The model is trained once on 1 January 2023 to 28 October 2025 (24,768 hours) and used unchanged for the whole test year.

Walk-forward retraining. The test year is divided into consecutive blocks of R days. Before each block, the model is retrained on all hours before the block start that fall inside the training window, and then forecasts the block. Nothing from inside a block is used to train the model that forecasts it. The main runs use $R = 14$ (27 retraining origins: 26 full blocks plus a final one-day block); $R = 7$ (53 origins: 52 full blocks plus a final one-day block) is used as a robustness check.

For walk-forward retraining, three kinds of training window are compared: *expanding* (always from 1 January 2023), and *rolling* windows of the last 365, 270, 180, 120 or 90 days. The rolling 180-day window is the main configuration. Because the window length was chosen on the test year, all window lengths are reported (section 4).

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3.6 Models

Baselines. *Naïve 24 h* repeats the price of the same hour on the previous day. *Naïve 168 h* repeats the price of the same hour one week earlier. The *weekday-aware naïve* common in the literature [5] uses the 168 h value for Mondays, Saturdays and Sundays and the 24 h value otherwise. *Hour mean 7 d* averages the same hour over the previous seven days.

Prophet. Prophet [19] with additive seasonality, Fourier orders of 10 (daily), 6 (weekly) and 10 (yearly), German public holidays, and a flat trend. A linear trend was tested and rejected because it extrapolated to negative price levels over the test year (section 4). Exogenous inputs enter as linear regressors.

Gradient boosting. Histogram-based gradient-boosted regression trees as implemented in scikit-learn [20] (`HistGradientBoostingRegressor`, following the binning approach of LightGBM [21]), with 500 iterations, learning rate 0.05, at most 31 leaves per tree, L2 regularisation 1.0, absolute-error loss and no early stopping, so every model is fitted with the full 500 iterations. The absolute-error loss makes the model target the conditional median, which matches the evaluation metric and reduces the influence of rare price spikes on the fit.

Ridge regression. A linear reference model on the same features, with median imputation, standardisation and the regularisation strength selected by efficient leave-one-out cross-validation over 20 log-spaced values between 10^{-2} and 10^{3}. The same grid is used for the single fit and for walk-forward retraining. At every refit, imputation, scaling and the regularisation strength are re-estimated on the current training window only, which contains only hours before the forecast origin; within that window the leave-one-out validation is not ordered in time.

3.7 Evaluation

The primary metric is the mean absolute error (MAE) in €/MWh over all 8,760 test hours; the root mean squared error (RMSE) is reported as a secondary metric that weights large errors more strongly. Percentage errors such as MAPE are not used, because prices close to zero and negative prices make them unstable or undefined.

Statistical significance. Differences between two models are tested with the Diebold–Mariano test [22]. Hourly errors within a day are strongly correlated, so the test uses the series of daily mean absolute-error differences ($n = 365$) with a Newey–West estimate of the long-run variance [23] with a lag of seven days, and a two-sided normal approximation.

Run-to-run variability. To assess run-to-run variability, the main configurations were refitted with five random seeds and early stopping enabled, so that a random validation split varies. None of these fits stopped early; the resulting MAE varied by at most ±0.3–0.5 €/MWh per configuration. Differences below this range are treated as noise. The models reported throughout the paper use the fixed 500-iteration schedule without early stopping, except for the three single-fit models without price lags in table 5, which come from earlier runs with scikit-learn's default early stopping.

Explanations. Feature contributions of the gradient-boosting models are computed with exact TreeSHAP [24, 25]. For walk-forward retraining, each test block is explained by the model that produced its forecasts, so the explanations can be compared across the year.

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3.8 Software and reproducibility

All experiments were run in Python 3.14.7 with NumPy 2.5.4, pandas 3.1.0, SciPy 1.18.1, scikit-learn 1.9.1, Prophet 1.5.0, holidays 0.107 and shap 0.54.1. The notebooks reproducing every table and figure, and scripts to download the input data from the original providers, are available at `https://doi.org/10.5281/zenodo.24082560`. Price data are not redistributed; they can be retrieved from the sources cited above.

4 Results

The results are presented in the order in which the experiments were run, because each step explains the next: the strength of the naïve baseline (section 4.1), the gain from realistic grid-operator forecasts under a single fit (section 4.2), the decay of that model over the test year and its repair by retraining (section 4.3), the value of perfect weather knowledge and the choice of algorithm once retraining is in place (section 4.4), and what the retrained models learned (section 4.5). Unless stated otherwise, all numbers refer to the full test year (8,760 hours) and are MAE in €/MWh.

4.1 Baselines and Prophet

Table 3 shows the day-ahead baselines. Repeating yesterday's price is the strongest of them (MAE 31.11), slightly ahead of the seven-day mean of the same hour. Repeating the same hour of the previous week is clearly worse (39.12).

Table 3: Day-ahead baselines on the test year.

Baseline	MAE	RMSE
Naïve 24 h	31.11	48.59
Hour mean 7 d	31.81	46.85
Weekday-aware naïve	33.22	51.99
Naïve 168 h	39.12	59.55

The weekday-aware naïve, a standard benchmark [5], loses to the plain 24 h naïve. Split by weekday, it loses almost entirely on Sundays (MAE 36.3 against 21.4), where it copies the previous Sunday instead of the preceding Saturday. On Mondays, where the plain naïve copies a Sunday, both are poor (42.0 and 43.0). In this market the state of the last one to three days, set by the current weather pattern and fuel price, carries more information than the weekday shape: a Saturday is a better guide to a Sunday than the Sunday a week earlier.

Prophet. Table 4 shows Prophet under different training windows and trends. With a linear trend, the model is very sensitive to the start of the training data: moving it from January to March 2023 increased the MAE from 54.6 to 92.3. The trend component, extended unchanged over 8,760 hours, fell from about 82 to 60 €/MWh across the test year with the January start, and from about 58 to −15 €/MWh with the March start, while 2026 prices averaged 112 €/MWh (table 1). With less than three years of data, the trend and the yearly seasonality are not well separated, and removing two expensive winter months shifted level from one to the other. A flat trend removes this failure mode, and training from 2023 is better than training on ten years.

No Prophet variant with realistic inputs beats the 24 h naïve. Adding yesterday's price and the previous-day mean as regressors brought the oracle-weather version to 29.11, but replacing the weather by the TSO forecasts gave 31.81. Removing only the residual load from that model restored 29.1, and neither a quadratic residual-load term nor additional weather regressors changed the result. The fitted coefficients explain why: without residual load, the weights of

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Table 4: Prophet on the test year. The weather regressors are observed (oracle) wind, solar and temperature.

		Prophet		Prophet + weather (oracle)	
Training data	Trend	MAE	RMSE	MAE	RMSE
From Jan 2023	linear	54.58	71.35	34.01	46.02
From Mar 2023	linear	92.33	108.77	39.18	53.38
From Jan 2023	flat	44.78	59.12	35.76	49.95
10 years	linear	42.95	57.28	37.73	51.37
10 years	flat	49.06	63.25	41.30	54.81

the two price-level regressors sum to about 0.32; with it, they fall to 0.22–0.23, and the wind and solar coefficients collapse (wind changes sign). The strongest physical input takes over part of the job of tracking the price level. Prophet therefore passes on only a third of a change in yesterday's price, and with residual load only a fifth; the naïve forecast passes on all of it. In a test year about 20 €/MWh more expensive than the training period, that difference decides the ranking.

4.2 Supervised models with a single fit

Table 5 shows gradient boosting trained once and used for the whole test year, with different information sets. Without price lags, no information set beats the naïve forecast: calendar features alone give 38.74, observed weather 34.14 and the TSO forecasts 37.03. With price lags alone the model reaches 28.05. The best realistic single-fit model combines lags with the TSO ratios (25.41, 18.3 % better than the naïve). Observed weather does better (23.35), and adding the TSO forecasts to observed weather makes the model worse (25.98 with ratios, 27.97 with absolute values).

Table 5: Single fit (trained once on 2023-01-01 to 2025-10-28); "oracle" marks observed weather. The three models without price lags were fitted with scikit-learn's default early stopping (section 3.7); all others with 500 iterations.

Information set	MAE	RMSE	vs naïve 24 h (%)
Gradient boosting, without price lags			
Calendar only	38.74	53.77	−24.5
TSO absolute	37.03	50.60	−19.0
Weather (oracle)	34.14	48.39	−9.7
Gradient boosting, with price lags			
Lags only	28.05	41.82	9.8
Lags + TSO absolute	27.63	41.15	11.2
Lags + TSO ratios	25.41	38.27	18.3
Lags + TSO ratios + weather (oracle)	25.98	39.60	16.5
Lags + weather (oracle)	23.35	36.20	24.9
Ridge regression, with price lags			
Lags + TSO ratios	23.57	35.83	24.2

Two explanations for the poor showing of the TSO forecasts were ruled out. Repeated fits with five random seeds varied by at most ±0.5, so the differences are not noise. And only 0.1–2.3 % of the test-year forecast values fall outside the range seen in training, so the trees are not extrapolating. The monthly errors point to the actual cause: the single-fit model was 50–60 %

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better than the naïve forecast from October to February, but worse than it from July onwards (fig. 1). The test year itself changes under the model.

4.3 Concept drift and walk-forward retraining

Table 6 shows the effect of retraining. For the model with lags only, retraining helps little (28.05 to 26.08 at best), and the expanding window, which uses the most data, is the best choice. For the model with lags and TSO ratios, retraining reduces the error by more than a third, from 25.41 to 15.98, and the shortest of the three windows wins. Price persistence is stable over the years, so more data helps; the relationship between the renewable share and the price changes quickly, so fresh data helps.

Table 6: Training scheme × information set, gradient boosting.

	Lags only		Lags + TSO ratios	
Scheme	MAE	RMSE	MAE	RMSE
Single fit	28.05	41.82	25.41	38.27
Retraining every 14 days				
Expanding (from 2023)	26.08	40.04	18.65	30.03
Rolling 365 d	27.09	41.19	17.94	28.97
Rolling 180 d	27.73	41.59	15.98	27.73
Retraining every 7 days				
Expanding (from 2023)	26.04	40.07	18.01	29.41
Rolling 365 d	27.46	41.48	17.20	28.38
Rolling 180 d	27.98	42.04	15.75	27.49

The retrained model (lags + TSO ratios, rolling 180 days, every 14 days) reaches an MAE of 15.98 and an RMSE of 27.73, 48.6 % better than the 24 h naïve. It beats the naïve forecast in every month of the test year by 40–67 % (fig. 1). In September and October 2026 all errors rise, because the daily price range widened to an average of up to 226 €/MWh per day, but the relative advantage stays near 50 %. The single fit, in contrast, reaches 59.6 in October 2026. In March and April the single fit is slightly better than the retrained model (16.3 against 16.6, and 14.7 against 16.6): a model trained on the 180 days before April has seen only autumn and winter, not a spring with strong solar generation.

Window length. Figure 2 shows the error for rolling windows from 90 to 365 days, with 95 % intervals from a moving-block bootstrap over days (blocks of seven days, 2,000 resamples). The intervals of the individual MAE values are wide (about ±1.7 €/MWh), because one test year contains a limited number of independent days, so the windows are compared with paired differences to the 180-day window, which remove the difficulty shared by all models on a given day. For 90, 120 and 270 days the paired difference is between 0.08 and 0.24, and every interval includes zero. For 365 days it is 1.96 [0.98, 3.05], and for the expanding window 2.67 [1.33, 3.95]. The curve is flat between 90 and 270 days and then steps up: data older than about nine to twelve months hurts, and within that limit the exact window length does not matter. With retraining every 7 days the pattern is the same (spread 0.14 between 90 and 270 days, step of 1.3 at 365 days). Because the window was chosen on the test year, this plateau matters for the credibility of the headline result: any window between 90 and 270 days gives an MAE between 15.7 and 16.2, 48–49.5 % better than the naïve forecast.

Retraining frequency. Retraining every 7 instead of every 14 days improves the MAE by about 0.2 (15.98 to 15.75), at twice the number of fits. The ranking of all configurations is un-

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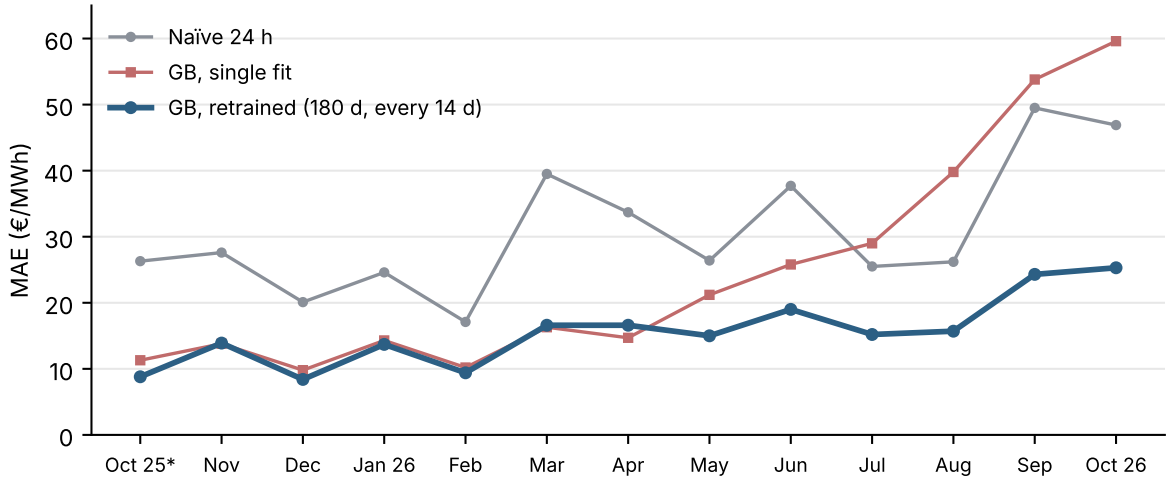


Figure 1: Monthly MAE over the test year. *October 2025 covers only 29–31 October.

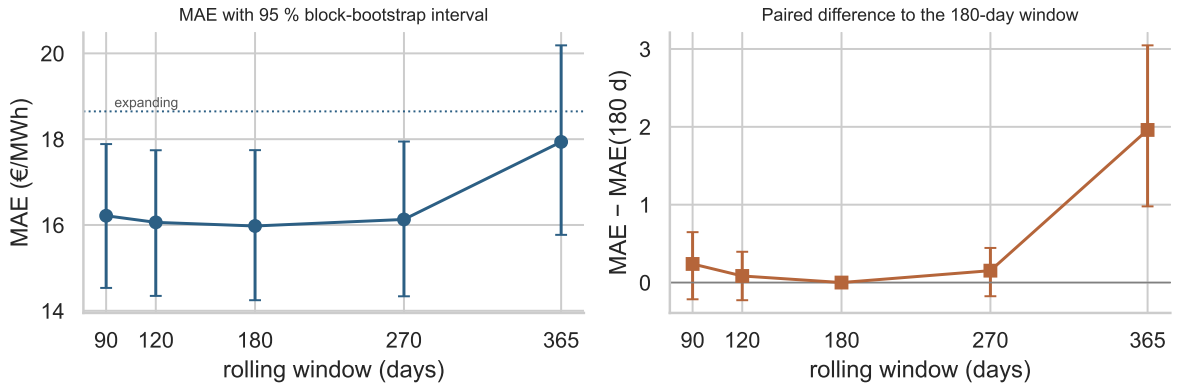


Figure 2: Rolling training window length (lags + TSO ratios, retraining every 14 days). Left: MAE with 95 % moving-block bootstrap intervals; dotted line: expanding window. Right: paired difference to the 180-day window with 95 % intervals.

changed. The decision to retrain, and the window length, matter much more than the frequency.

Significance. Table 7 shows Diebold–Mariano tests for the main comparisons. In each row, model A is the one with the lower MAE, so every difference is negative and the question is only whether A's advantage is significant. The retrained model is significantly better than the naïve forecast, than its own single-fit version, than the same scheme without TSO forecasts, than Ridge regression with the same inputs and retraining, and than the retrained model with observed weather instead of TSO forecasts ($p < 0.001$ in all cases). With retraining every 7 days the corresponding differences are −15.36, −9.65, −12.23, −4.09 and −5.01 €/MWh, with DM statistics between −5.0 and −11.7 (all $p < 0.001$). The verdicts do not depend on the Newey–West lag: with lags of 3 and 14 days the DM statistics move by at most 1.7 (for example −6.67 and −3.79 for the comparison with the single fit, against −4.96 at lag 7), and every p-value stays on the same side of 0.05.

Table 7: Diebold–Mariano tests on daily mean absolute errors (Newey–West, lag 7), retraining every 14 days. Model A is always the model with the lower MAE; Δ = MAE(A) − MAE(B), which equals the mean daily difference, so a negative value means A is better.

Model B	MAE B	Δ	DM	p
A = GB lags + TSO ratios, retrained 180 d (MAE 15.98)				
Naïve 24 h	31.11	−15.13	−11.23	< 0.001
Same model, single fit	25.41	−9.43	−4.96	< 0.001
GB lags only, retrained 180 d	27.73	−11.75	−11.83	< 0.001
Ridge, same inputs, retrained 180 d	19.98	−4.01	−7.49	< 0.001
GB lags + weather (oracle), retrained 180 d	20.80	−4.83	−8.05	< 0.001
A = GB lags + TSO ratios + weather (oracle), retrained 180 d (MAE 15.82)				
GB lags + TSO ratios, retrained 180 d	15.98	−0.16	−0.96	0.34

4.4 Perfect weather knowledge and the choice of algorithm

Retraining reverses the single-fit ranking of observed weather and TSO forecasts (table 8). With retraining, the realistic model with TSO ratios (15.98) is significantly better than the model with observed weather (20.80). Adding observed weather on top of the TSO forecasts changes the error by −0.16, which is not significant ($p = 0.34$, table 7). With retraining every 7 days the difference is −0.30 (15.46 against 15.75), which is marginally significant ($p = 0.04$ for Newey–West lags of 3, 7 and 14 days). With five model comparisons and two retraining frequencies, a Bonferroni-type correction would push this result above the conventional 0.05 threshold; we therefore treat it as suggestive rather than robust. Once the model is kept up to date, perfect hindsight about the weather is worth at most about 0.3 €/MWh, or 2 % of the error, on top of the grid operators' day-ahead forecasts. The single-fit advantage of observed weather was a drift effect: the models that rely on the TSO forecasts lost most when they were not updated. The same holds for absolute values versus ratios. Their gap shrinks from 2.2 to 0.3 with retraining every 14 days and disappears with retraining every 7 days (15.71 against 15.75). Observed weather gains much less from retraining (2.6) than the TSO forecasts (9–11). A likely reason is that the TSO forecasts already include the installed capacity, which grows every year, whereas a model given wind speed and irradiance would have to learn the size of the fleet from 180 days of data.

Ridge versus gradient boosting. With a single fit, Ridge regression is better than gradient boosting on the same inputs (23.57 against 25.41): a linear model degrades more slowly when

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Table 8: Single fit versus retraining (rolling 180 d, every 14 d). Gain = reduction in MAE.

	Single fit		Retrained		
Model	MAE	RMSE	MAE	RMSE	Gain
GB lags + TSO ratios + weather (oracle)	25.98	39.60	15.82	27.60	10.16
GB lags + TSO ratios	25.41	38.27	15.98	27.73	9.43
GB lags + TSO absolute	27.63	41.15	16.29	27.59	11.34
GB lags + weather (oracle)	23.35	36.20	20.80	33.73	2.55
GB lags	28.05	41.82	27.73	41.59	0.33
Ridge, lags + TSO ratios	23.57	35.83	19.98	30.80	3.59

the relationships drift. With retraining, gradient boosting is significantly better (15.98 against 19.98, DM = −7.49). When the model is kept current, its ability to represent the curved merit-order relationship pays off. Which algorithm is better therefore depends on the training scheme.

4.5 What the retrained models learned

Table 9 summarises the TreeSHAP contributions by feature group. In all models the TSO ratios carry the largest share. Compared with the single fit, the retrained models rely less on the price lags (18.7 against 24.9) and more on the calendar (15.0 against 10.7). A model trained on the last 180 days already sits at the current price level, so part of the level information moves from the lag features into the training data. And with 180 days of data, the hour-of-day pattern the model learns belongs to the current season, while the single fit averages a daily shape over three years in which the midday price dip deepened considerably. With a 90-day window the calendar weight barely rises (11.3), because three months of data contain too few months to learn seasonal patterns, and the model relies more on the forecasts instead. The 90- and 180-day models have nearly the same error but distribute their reliance differently.

Table 9: Mean absolute SHAP contribution by feature group (€/MWh), summed over the features of each group. Group values are comparable with each other but do not add up to a forecast.

Group	Retrained, 180 d	Retrained, 90 d	Single fit
TSO forecast ratios	35.7	37.0	34.2
Price level (lags)	18.7	17.7	24.9
Calendar	15.0	11.3	10.7

Importance through the year. Figure 3 shows how the explanation of the retrained model is divided between the groups in each month. In the solar season (April to July) the TSO ratios account for about 60 %: how deep the midday price falls depends on tomorrow's solar generation, which changes from day to day with cloud cover. In winter the price lags reach their highest share (about 30 %): prices are mostly set by gas plants, and wind conditions persist for several days with weather fronts, so the price level carries over. From August to October the calendar share rises to about 30 %, as the retrained models learn the extreme daily shape of late summer 2026, with near-zero middays and evening peaks. This is the mechanism behind the failure of the single fit. It did not lack information; the meaning of its inputs changed, and it had no way to learn that.

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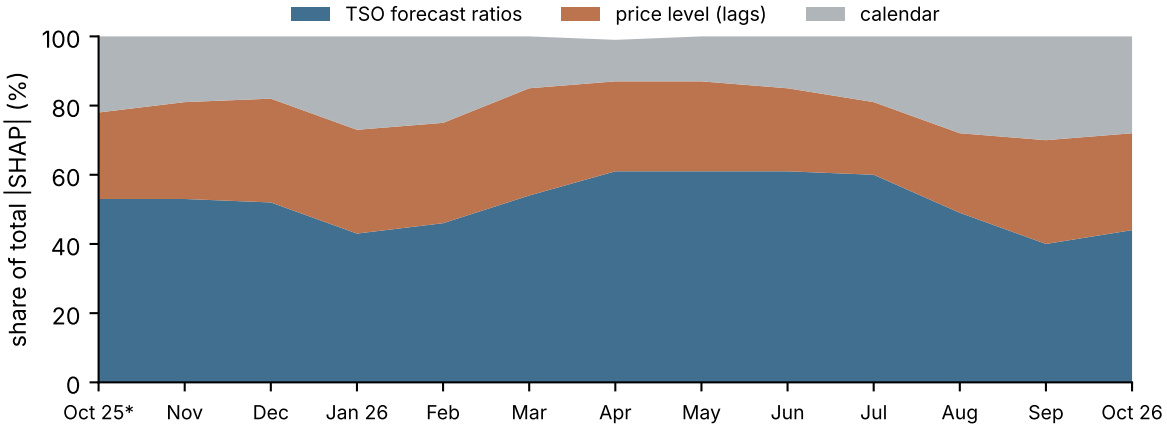


Figure 3: Share of the total absolute SHAP contribution per feature group, by month (retrained, rolling 180 d, every 14 d). *October 2025 covers only 29–31 October.

The learned merit order. Figure 4 shows the residual-load effect, the summed contribution of s^{RL} and s^{RE}, as a function of the residual-load share, separately for test hours forecast by models trained in winter (November to January) and in summer (July to September). The curve has the shape of a merit order: a flat floor when renewables cover the whole load, a steep section between shares of about 0.05 and 0.4 where small changes decide whether thermal plants are needed, and a slower rise above that. The two seasons differ in three ways. The summer floor is deeper (−106 against −82 €/MWh) and extends to negative shares down to −0.36, which winter hours never reach. The summer curve lies 6–10 €/MWh higher at shares above 0.5. And its total range is larger (about 150 against 120 €/MWh). The same input is worth more in summer 2026 than in winter, which is the relationship a single fit cannot follow. The point where a curve crosses zero marks the typical share for its training period, not a price threshold, since SHAP contributions are measured relative to the model's average forecast.

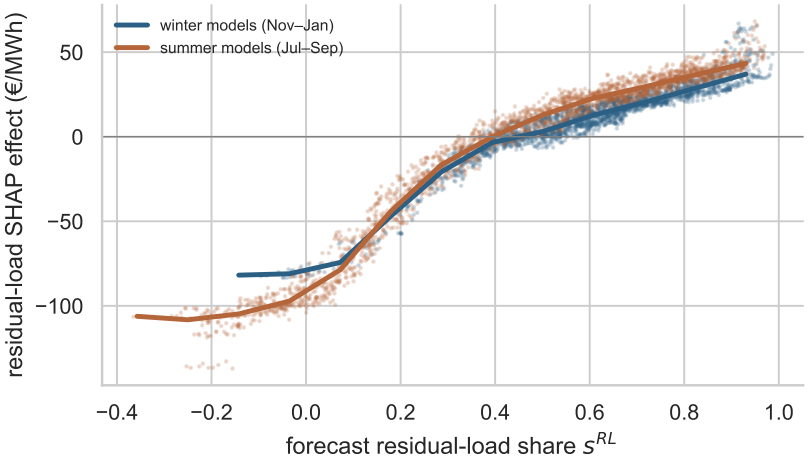


Figure 4: Residual-load effect (sum of the TreeSHAP contributions of s^{RL} and s^{RE}) against the forecast residual-load share, for test hours forecast by winter models (November–January) and summer models (July–September). Lines: binned medians.

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Extreme hours. The two most extreme hours of the test year show the limits of the model. For the highest price, on Monday 14 September 2026 at 17:00 UTC (697 €/MWh), the model forecast 244 €/MWh, starting from its average of 103 €/MWh and driven by the residual-load effect (+51), the seven-day hour mean (+25) and yesterday's price (+16). For the lowest price, on Friday 1 May 2026 at 11:00 UTC, a public holiday (−499 €/MWh as an hourly average, practically at the harmonised minimum clearing price of −500 €/MWh in force at the time [26, 27])[1], the model forecast −84 €/MWh, starting from its average of 97 €/MWh and driven by the residual-load effect (−124), the solar share (−27) and yesterday's price (−15). The direction and the drivers are right, but the magnitude is a third or less of the actual value. The hourly average of −499 €/MWh also implies that the four underlying quarter-hour prices were all within 4 €/MWh of the −500 €/MWh floor, which is consistent with the hourly series being an average of quarter-hour prices (sections 3.1 and 5.5). The model's shortfall at these hours is largely a consequence of the absolute-error loss, which targets the conditional median and therefore does not chase rare extremes. For typical hours this is what the MAE rewards, including most negative-price hours, where the error of a median forecast stays moderate. Forecasting the extremes themselves is a different task, outside the scope of this study; quantile losses for upper and lower price quantiles, or a two-stage approach that first classifies spike and negative-price hours and then estimates their size, are the natural tools for it (section 5).

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5 Discussion

5.1 Why the single fit failed: concept shift, not covariate shift

A model can degrade over time for two different reasons. Under *covariate shift*, the inputs take values the model has rarely seen. Under *concept shift*, the inputs look familiar, but the price responds to them differently. To tell them apart, three measures were computed for each test month and compared with the excess error of the single fit over the retrained model (fig. 5):

- the population stability index (PSI) of each TSO ratio, comparing the month with the single fit's training data (covariate shift);
- the *merit-order shift*: the mean absolute difference between the month's average price and the training average price within the same residual-load-share bin, using ten bins from the training distribution (concept shift);
- the *daily-shape shift*: the mean absolute difference between the month's average price for each hour of the day and the training average, after removing each profile's own mean (concept shift).

The merit-order shift tracks the excess error closely (Spearman $\rho = 0.83$, $n = 12$, approximate 95 % interval [0.48, 0.95]), rising from 3–20 €/MWh in winter to 56–84 €/MWh in August to October 2026, when the single fit's excess error reached 24–34 €/MWh. The covariate measure does not: for the largest PSI of the four ratios, $\rho = -0.06$ (approximate interval $[-0.64, 0.57]$), indistinguishable from zero. The covariate shift is in fact largest in winter, with PSI values for the solar share of 0.77–1.0 between November and January, far above the usual threshold of 0.25 for a large shift [29], yet those are the months in which the single fit did well. A shift the model has seen before, such as the seasonal swing in solar generation that is present in every training year, does no harm. What harmed the single fit was that the price paid at a given residual-load

[1] Prices below 70 % of the minimum on two days within 30 days trigger an automatic lowering of the minimum by 100 €/MWh. Delivery day 1 May 2026 was the second of the two trigger days, and the minimum was set to −600 €/MWh from the auction of 28 May 2026 [28].

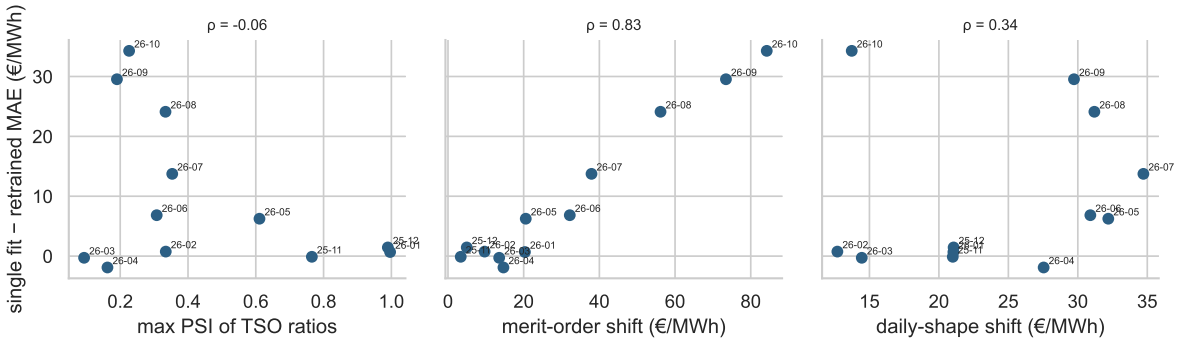


Figure 5: Excess MAE of the single fit over the retrained model, per test month, against three drift measures; ρ is the Spearman rank correlation over 12 months (October 2025, with 3 days, is excluded). Labels give year and month.

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share rose, by as much as 84 €/MWh in October 2026. This is consistent with the range check in section 4.2, which found almost no test values outside the training range.

The merit-order shift is closely related to the overall price-level shift ($\rho = 0.73$ with the excess error), because a merit order that moves up raises prices at every share. Its likely drivers, fuel and carbon prices, are not among the model's inputs. Within a 180-day window the model learns the current position of the merit order from the data, which is why retraining repairs what a single fit cannot. The daily-shape shift correlates more weakly ($\rho = 0.34$); it is high from April to September, when the midday price dip is deepest, and contributes to the excess error in early summer. With 12 months the correlations are descriptive rather than formal tests, but the separation between the measures is clear. The intervals are Fisher-z approximations with the usual variance adjustment for Spearman's ρ.

5.2 The training scheme decides the ranking

Three comparisons in this study change sign between a single fit and walk-forward retraining. Observed weather beats the TSO forecasts under a single fit but loses clearly with retraining. Absolute TSO values lose to ratios under a single fit but match them with retraining. And Ridge regression beats gradient boosting under a single fit but loses significantly with retraining. In each case the single fit rewards robustness to drift, and retraining rewards the ability to use the information. A comparison of models or input sets on a market with this much drift therefore says as much about the evaluation protocol as about the models. Studies that train once and test over a long period, as for example in [9], may rank models differently from the way they would perform in operation.

The same reasoning explains why the 24 h naïve forecast is hard to beat. It refreshes the price level every day, which is the property the single-fit models lacked. Prophet, which adds its inputs with fixed weights, passes on only a fifth to a third of a level change and could not compete with realistic inputs. The supervised model only beat the naïve forecast by a wide margin once it was given the level through lagged prices and kept up to date through retraining.

5.3 Practical recommendations

For day-ahead forecasting in the German market under conditions like those of 2025–2026, the results suggest:

- Retrain regularly. Every one to two weeks was sufficient here; retraining every week instead of every two weeks improved the MAE by only about 0.2 €/MWh.

- Train on the last three to nine months. Longer windows hurt, and within that range the length matters little.
- Use the TSO day-ahead forecasts, expressed as shares of the forecast load, together with lagged prices. With retraining, perfect knowledge of the weather added at most about 0.3 €/MWh on top of them, so a forecaster does not need a weather model of their own.
- Choose the algorithm after the training scheme. With retraining, gradient boosting was significantly better than Ridge regression; without it, the opposite held.

5.4 Relation to the research question

The study asked whether time-series models or supervised models are better suited to German day-ahead prices, which are driven largely by weather and residual load. Neither is sufficient on its own. The pure time-series models either ignore the physical drivers (naïve forecasts) or cannot combine them with the price level (Prophet). The supervised model without price lags never beats the naïve forecast, and with lags but without retraining it decays within months. The model that wins is supervised, but it needs time-series memory in its inputs and is trained in a time-series way. The combination is what produces the 48.6 % improvement over the naïve forecast.

5.5 Limitations

- **One market, one test year.** All results refer to the DE-LU bidding zone and the year from October 2025 to October 2026. The rolling evaluation covers 27 retraining origins within that year (26 full 14-day blocks plus a final partial block), but other years, with different fuel prices and weather, may favour other window lengths.
- **Window chosen on the test year.** The 180-day window was selected by looking at test results. The plateau between 90 and 270 days, confirmed by paired bootstrap intervals, limits the effect of this choice: the paired difference to the 180-day window is at most 0.24 €/MWh for all windows between 90 and 270 days.
- **Publication time of the TSO forecasts.** The forecasts are formally due by 18:00 on the day before delivery, after the noon auction (section 3.2). If a forecaster only had older forecasts at gate closure, the results would be somewhat optimistic.
- **15-minute products.** Since delivery day 1 October 2025, the European day-ahead auction clears 15-minute products [30], so the whole test year falls under the new market design. EPEX SPOT publishes a 60-minute price index as the arithmetic average of the four 15-minute clearing prices [14], and the hourly prices used here most likely correspond to this index. Price movements within the hour are therefore averaged out: the model forecasts hourly averages and cannot use, or be judged on, quarter-hour variation. A 15-minute target would be more volatile and may favour different features.
- **Missing drivers.** Fuel and carbon prices, which the drift analysis points to as the likely cause of the merit-order shift, are not used. Neither are cross-border flows or storage behaviour.
- **Point forecasts of the median.** The absolute-error loss targets the conditional median. The model gets the direction of extreme hours right but captures only a third or less of their size, and it produces no uncertainty estimates.
- **Single price source.** Prices were taken from one provider and not cross-checked against a second source.

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5.6 Outlook

Four extensions follow directly from these results. Fuel and carbon prices as inputs could explain part of the merit-order shift that retraining currently has to absorb. Exponentially decaying sample weights, instead of a hard window cutoff, could keep the seasonal memory that the 180-day window loses in spring while still favouring recent data. Quantile losses, or a two-stage model that first classifies spike and negative-price hours and then estimates their size, would address the extreme hours and provide uncertainty estimates. And repeating the analysis on further test years, on the 15-minute target and on neighbouring bidding zones would show how general the window plateau and the reversal of rankings are.

6 Conclusions

On German day-ahead prices from October 2025 to October 2026, a gradient-boosting model with lagged prices and the TSO day-ahead forecasts of load, wind and solar, expressed as shares of the forecast load and retrained every two weeks on the last 180 days, reached an MAE of 15.98 €/MWh. That is 48.6 % better than repeating yesterday's price, significantly better ($p < 0.001$), and better in every month of the test year. Any training window between 90 and 270 days gives practically the same result, so the result does not depend on the window that was chosen.

The same model, trained once and used for the whole year, was only 18 % better than the naïve forecast and became worse than it from July 2026, because the price paid at a given residual load moved upward while the inputs themselves stayed within familiar ranges. Retraining did more than repair this decay: it changed which models and inputs looked best. Under a single fit, observed weather, absolute TSO forecasts and Ridge regression came out ahead; with retraining, all three lost. Perfect knowledge of the weather was then worth at most about 0.3 €/MWh on top of the grid operators' forecasts.

The ranking of algorithms and input sets is therefore not fixed: which model is best depends on the training scheme, and the training scheme itself is the first-order decision. The answer to the question of time series versus supervised learning is neither. A supervised model needs time-series memory in its inputs and time-series practice in its training to use the physical information it is given. For forecasting markets that change as fast as the German one, how a model is kept up to date matters more than which model it is.

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